

SKHASH: A Python Package for Computing Earthquake Focal Mechanisms

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Abstract

We introduce a Python package for computing focal mechanism solutions. This algorithm, which we refer to as SKHASH, is largely based on the HASH algorithm originally written in Fortran over 20 yr ago. HASH innovated the use of suites of solutions, spanning the expected errors in polarities and takeoff angles, to estimate focal mechanism uncertainty. SKHASH benefits from new features with flexible input formats and allows users to take advantage of recent advances in constraining focal mechanisms for small magnitude or poorly recorded earthquakes. The 3D locations of earthquakes and the velocity models used are varied when finding acceptable solutions. As a result, source–receiver azimuths are reflective of errors from the earthquake locations and velocity models, in addition to the takeoff angles. Users can consider weighted *P*-wave first-motion polarities derived from traditional or machine-learning picks, cross-correlation consensus, and/or imputation techniques using SKHASH. Focal mechanism solutions can also be further constrained using traditional, machine learning, and/or cross-correlation consensus *S/P* amplitude ratios. With improved reporting of individual and collective *P* polarity and *S/P* amplitude misfits, users can better evaluate the success of the solutions and the quality of the measurements. The reporting also makes it easier to identify potential issues with metadata, including incorrectly reported station polarity reversals. In addition, by leveraging vectorized operations, taking advantage of an efficient backend Python C Application Programming Interface, and the use of a parallel environment, the Python SKHASH routine may compute mechanisms quicker than the HASH routine.

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Introduction

Earthquake focal mechanisms can provide information about fault structures, rupture kinematics, and stress orientations. We commonly determine an earthquake's focal mechanism by analyzing energy recorded across a seismic network as the radiation pattern of *P* and *S* waves from an earthquake depend on the orientation of the fault and how it slipped (i.e., strike-slip, normal, reverse, or oblique). Assuming a double-couple model, we can imagine a four-quadrant sphere encompassing the point source. Two of these quadrants will have initial compressional movement away from the source, and the other two quadrants will have dilatational movement toward the source. By determining the *P*-wave first motions recorded on seismometers and the paths the energy traveled to each instrument, we can identify the quadrant orientations that fit our observations. We can also constrain the potential quadrant orientations by considering the relative amplitudes of the *P* and *S* waves recorded on the seismometers. The amplitudes of *P* waves are largest near the compressional and tensional axes while *S* waves are largest near the nodal planes. By looking

for quadrant orientations that match this amplitude pattern in addition to the previously described *P*-wave first-motion polarities, we can be more confident in how the quadrants are oriented. The resulting solution that describes the quadrant orientation is referred to as a focal mechanism.

When an earthquake is well recorded and an accurate velocity model can be determined, we can extract more information about how the fault slipped and better interpret non-double-couple sources. Moment tensor inversions can be performed for larger (commonly $M > 3.5$ – 4.0) earthquakes recorded by regional seismic networks by comparing Green's functions with observed data (e.g., Guilhem and Dreger, 2011). Similar approaches can also be used to constrain the nodal planes for smaller magnitude seismicity if the events are recorded on a

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local seismic network (e.g., [Li et al., 2011](#)). However, for the ubiquitous $M < 4.0$ earthquakes that do not occur near a local network, dependable moment tensors often cannot be produced. For these smaller magnitude (or otherwise poorly recorded) earthquakes, we may instead assume a double-couple source and compute focal mechanism solutions.

Although we have been largely limited to using individually determined P -wave first-motion polarities and S/P amplitude ratios to compute focal mechanisms in the past, recent advancements have improved our ability to determine these features. These advancements include making cross-correlation relative polarity measurements (e.g., [Shelly et al., 2016](#)), using machine learning to detect additional phases (e.g., [Uchide, 2020](#); [Cheng et al., 2023](#)) or impute estimates for low confidence or missing measurements (e.g., [Skoumal, Shelly, and Hardebeck, 2023](#)), using Bayesian source inversion considering measurement uncertainties (e.g., [Pugh et al., 2016](#)), making S/P ratio measurements on single-component instruments ([Shelly et al., 2022](#)), and making cross-correlation relative S/P ratio measurements ([Skoumal, Hardebeck, and Shelly, 2023](#)). These techniques have allowed focal mechanisms to be determined for even smaller magnitude or poorly recorded earthquakes, allowing for greater insight into complex fault geometries and behaviors (e.g., [Shelly et al., 2023](#)).

The FPFIT ([Reasenbergs and Oppenheimer, 1985](#)) and HASH ([Hardebeck and Shearer, 2002, 2003](#)) software packages are commonly used to determine focal mechanisms. Both algorithms use a grid search considering the possible strike, dip, and rake solutions to identify the focal mechanism that best describes the polarities. The HASH algorithm has additional features, including the abilities to consider uncertainties in takeoff angles by varying the hypocentral depths and velocity models that are considered. The FPFIT and HASH algorithms were written in Fortran and are computationally efficient, but they do not include recent advancements in focal mechanism determination. HASHpy ([Williams, 2014](#)) allows users to call HASH subroutines using a Python wrapper, but the previously mentioned limitations of the algorithm still apply.

Here, we introduce the SKHASH algorithm ([Skoumal et al., 2024](#); see [Data and Resources](#)) written in Python. The primary objectives of this algorithm are to make the computation of focal mechanisms as easy and quick as possible for scientists while also allowing users to take advantage of recent advancements in generating focal mechanism solutions. Only two freely available, widely used external Python packages are required for SKHASH: NumPy ([Harris et al., 2020](#)) and pandas ([McKinney, 2010](#)). These are among the most popular Python packages ([PyPI Stats, 2024](#)) and are included by default in the popular Anaconda Python distribution ([Anaconda Software Distribution, 2016](#)). SKHASH can run on Linux, macOS, and Windows using all currently supported versions of Python (as of February 2024: v.3.8–3.12).

Method

The SKHASH algorithm (Fig. 1) is largely based on HASH ([Hardebeck and Shearer, 2002; 2003](#)). Both algorithms use a grid-search approach that identifies the set of acceptable focal mechanism solutions given the inputted P polarity and S/P ratio measurements. The grid search can be repeated considering differing combinations of variable inputs (e.g., hypocentral depths) to produce suites of solutions. The variability of this resulting set of acceptable models defines the uncertainty of the preferred focal mechanism solutions. A primary objective of SKHASH is to make computing focal mechanisms easier for a variety of workflows. SKHASH has additional capabilities and features that improve the ease of use, some of which we will describe here. These include facilitating a variety of input formats, more friendly error messaging, and automatic quality control reporting. The SKHASH algorithm was also created with recent advancements (e.g., consensus polarities) and machine-learning workflows in mind, allowing the results from these new techniques to be easily incorporated into the inversion.

New features

Three principal inputs are necessary to compute a focal mechanism: (1) takeoff angles; (2) source–receiver azimuths; and (3) the corresponding observations (i.e., P -wave polarity and/or S/P ratio measurements). As stated previously, given vertical depth uncertainties and velocity models provided by the user, HASH varies the hypocentral depths of earthquakes and velocity models considered to compute suites of takeoff angles. For source–receiver azimuths, HASH computes the azimuth between the station and preferred epicentral location and does not consider horizontal location uncertainties (Fig. 2a). With SKHASH, the hypocentral location is varied in three dimensions, which allows for variable takeoff angles and source–receiver azimuths to be considered (Fig. 2b). If the azimuthal and/or the takeoff uncertainty for a measurement exceeds a value set by the user, the measurement can be automatically discarded. The degree of benefit of considering horizontal errors will depend on the source–receiver distances and horizontal uncertainties. Large source–receiver distances and/or negligible horizontal hypocentral uncertainties will yield small variations in source–receiver azimuths, whereas instruments close to the receiver and/or large uncertainties (Fig. 2b) will result in larger azimuthal variations.

We add the functionality for custom weights to be used for polarity measurements. This gives users much greater control over the significance that individual polarity measurements have on determining the resulting solution. The significance of a measurement is denoted by the absolute value of the polarity value ranging between -1.0 and 1.0 while the sign denotes the upward (compressive) or downward (dilatational) movement.

The user has the option to automatically report misfits for polarities and S/P ratios (Fig. 3). These can be reported for individual measurements or as a collective for each receiver.

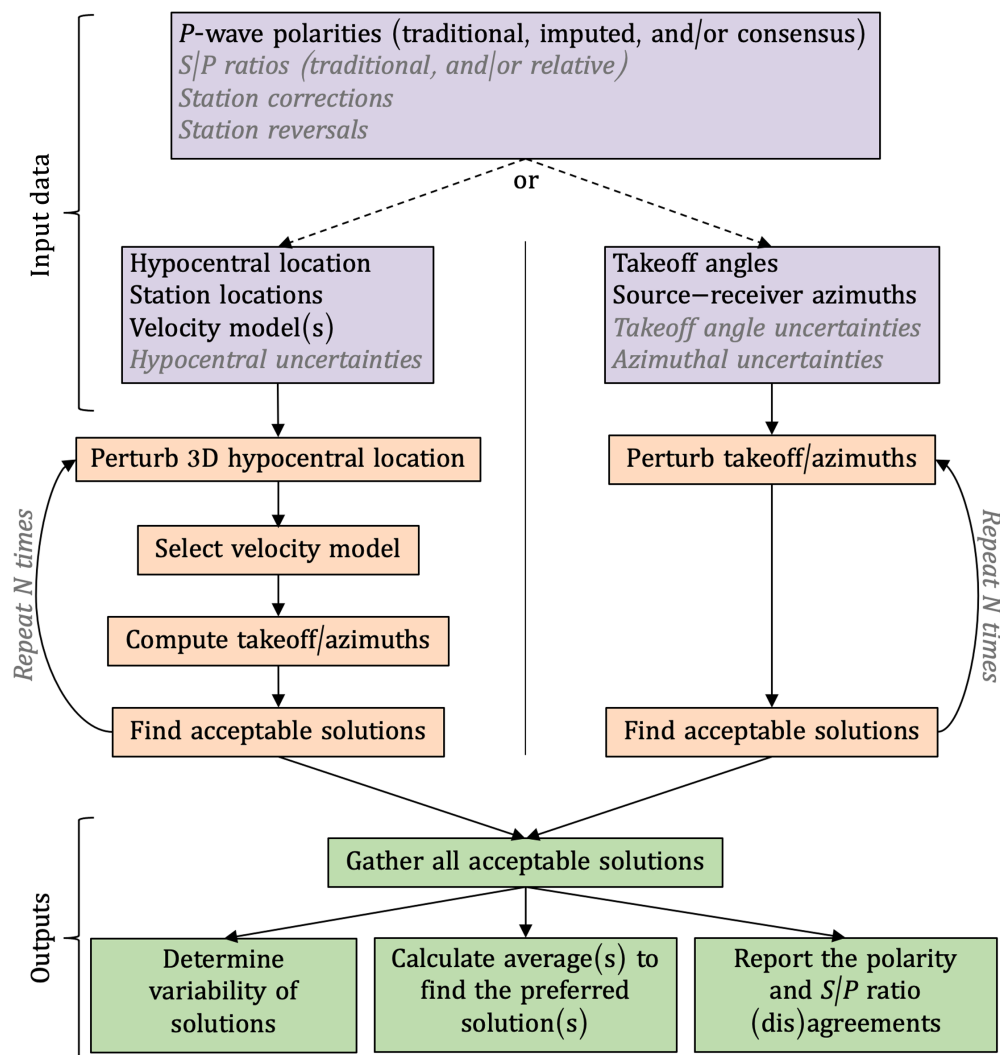


Figure 1. Algorithm flowchart of the modified HASH method (Hardebeck and Shearer, 2002, 2003) used in SKHASH (Skoumal et al., 2024). Italicized, gray text designates optional (but recommend) inputs and steps. The color version of this figure is available only in the electronic edition.

This reporting also makes it easier to identify potential issues with the data and/or metadata, including incorrectly reported station polarity reversals.

Usually, the polarity measurements recorded for a given earthquake are used to compute a focal mechanism solution for the event. Alternatively, measurements from multiple, similar earthquakes can be used together to produce composite mechanisms (e.g., Shelly et al., 2016). With the composite mechanism approach, many earthquakes can contribute to a single solution that represents the “family” of earthquakes. This can help address the limitation of poor focal sphere coverage that is common for poorly recorded events. SKHASH is capable of computing both single-event and composite focal mechanism solutions (Fig. 4).

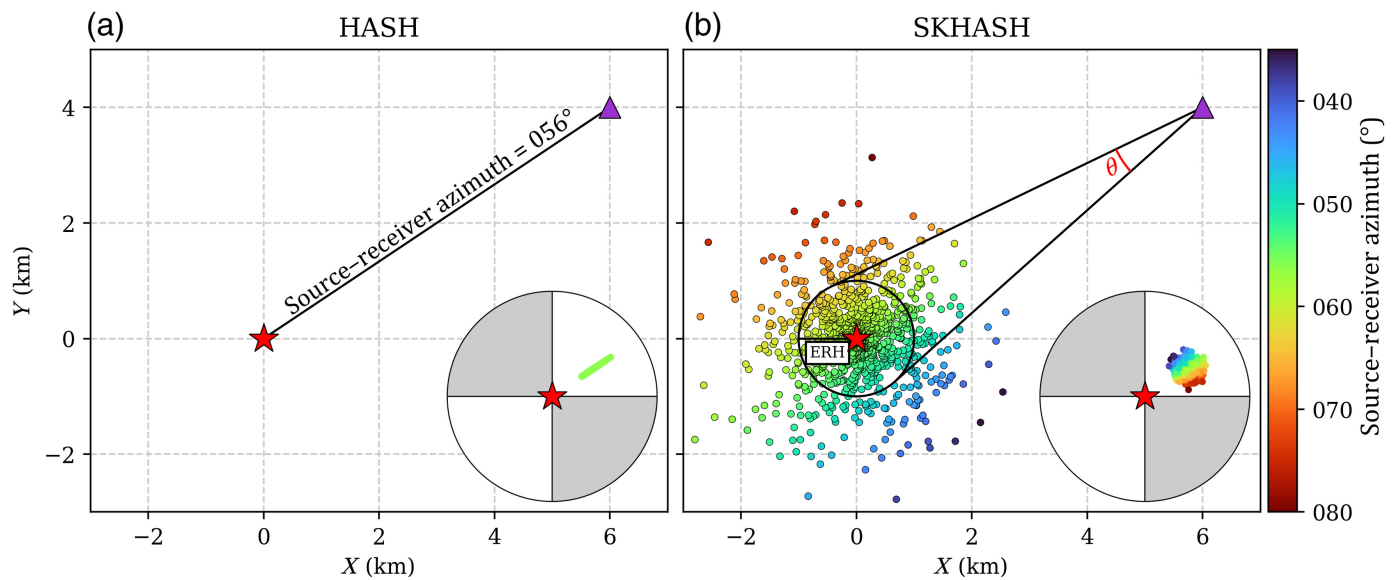
Recent advancements using cross correlation and machine learning have improved the ability to constrain focal mechanisms,

and SKHASH is designed to accommodate results from these new techniques. Users have the option of using any combination of (1) traditional *P*-wave first motions, (2) polarities identified with deep learning (e.g., Uchide, 2020), (3) correlation consensus polarities, which use cross correlation of phase arrivals to identify (anti)similar waveforms (e.g., Shelly et al., 2016), and (4) imputed polarities, which fill in missing or low-confidence measurements using machine-learning techniques (e.g., Skoumal, Shelly, and Hardebeck, 2023). In addition, any combination of traditional and relative *S/P* ratio measurements derived from single-component (Shelly et al., 2022) and multi-component stations (Skoumal, Hardebeck, and Shelly, 2023) can also be used. The measurement sources are tracked, and the accuracy of each measurement type can be reported to help users gauge the contribution from the different measurement sources. These reports can be automatically produced for individual measurements as well as collectively reported for each receiver.

SKHASH offers better control over the relationship between station metadata and the polarity and *S/P* ratio measurements. For example, when using HASH, polarity reversals are applied to all instruments that share the station code within a specified time window. With SKHASH, instrument locations, reversals, and station correction measurements can be related using any combination of the network, station, location, and channel code metadata at multiple time windows. Although we recommend using all network, station, location, and channel codes to associate measurements with the metadata whenever possible, being able to omit some metadata can still be useful. For example, old HYPOINVERSE (Klein, 1989) phase files do not include location codes.

Ease of use

With HASH, different versions of the Fortran code (“drivers”) are compiled by the user depending on the format of input data



and type of analysis performed (e.g., whether S/P ratios are included). Summarized briefly, the five drivers included in HASH v.1.2 handle (1) P -wave polarities and precomputed takeoff angles/azimuths, (2) P -wave polarities and 1D velocity model(s) to compute the takeoff angles/azimuths, (3) P -wave polarities and S/P ratios and velocity model(s) to compute the takeoff angles/azimuths, (4) similar to (2) but with an updated Southern California Earthquake Data Center phase format, and (5) similar to (1) but using the takeoff angles/azimuths from 3D ray tracing reported in the output file from SIMULPS 3D (Evans *et al.*, 1994). Users may create custom

Figure 2. Demonstration of source-receiver azimuthal calculations using (a) HASH (Hardebeck and Shearer, 2002, 2003) and (b) SKHASH (Skoumal *et al.*, 2024). With HASH, the takeoff angles are varied but a single source-receiver azimuth is computed between each epicenter (star) and receiver (triangle). With SKHASH, in addition to varying the takeoff angles, the epicentral location for each event can be varied given a horizontal error (ERH) that results in a more representative reflection of the potential source-receiver azimuths. The user has the option to automatically discard measurements that exceed a given source-receiver azimuthal uncertainty (θ). The color version of this figure is available only in the electronic edition.

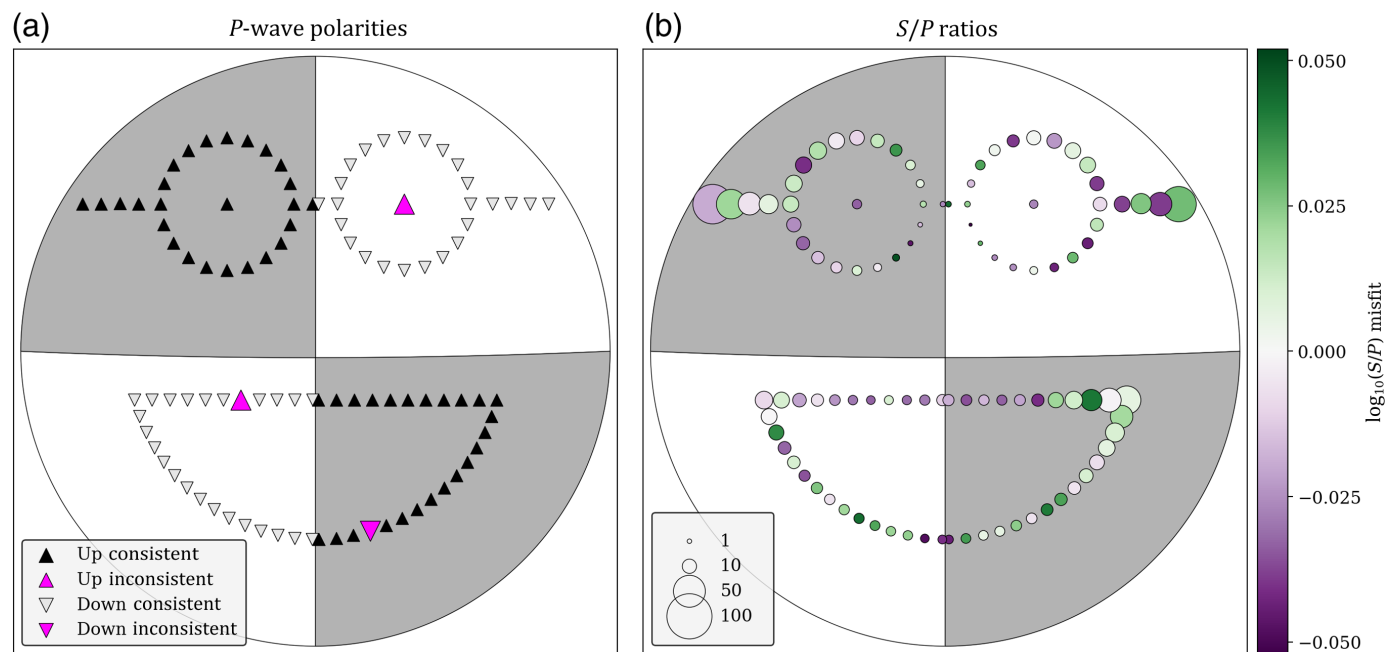
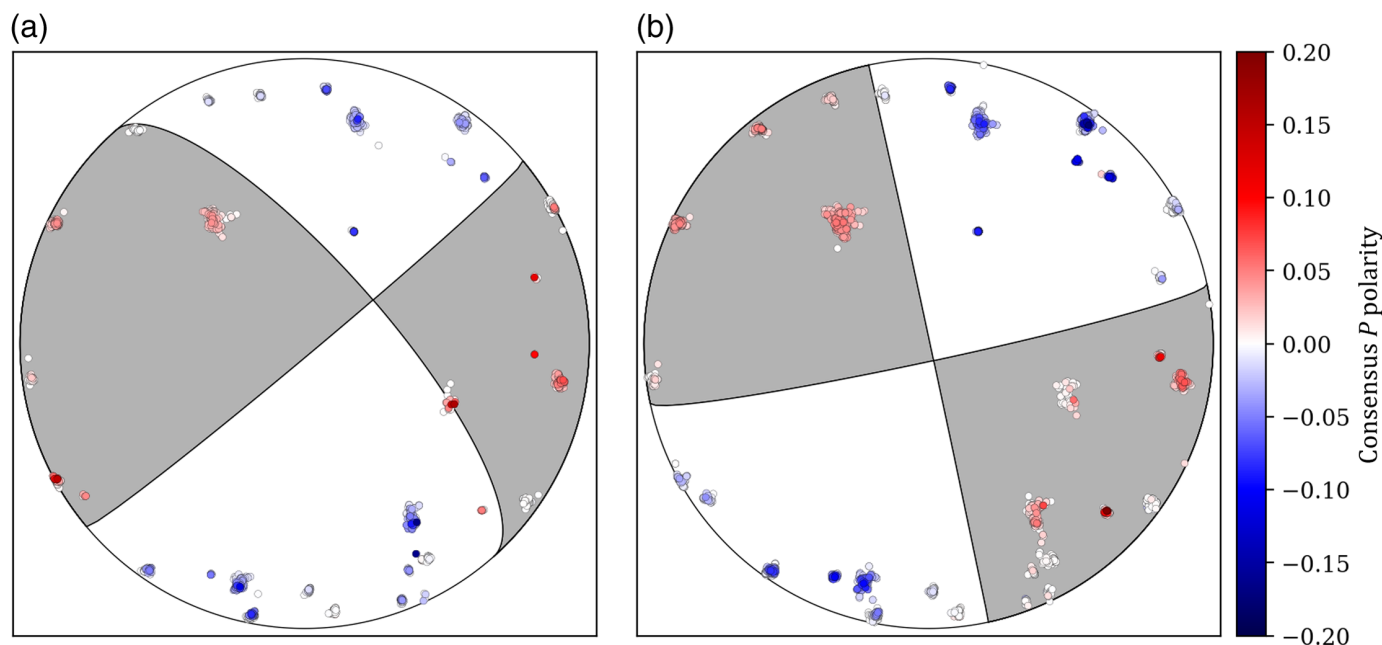


Figure 3. Demonstration of the misfit reporting using the SKHASH (Skoumal *et al.*, 2024) example “smile” composed of synthetic

(a) P -wave polarities and (b) S/P ratios. The color version of this figure is available only in the electronic edition.



HASH drivers by modifying the existing code, but it requires some level of Fortran proficiency and familiarity with the algorithm.

With SKHASH, formatting of the user input files is more flexible and handles varying inputs without the need of compiling different versions of the code. SKHASH is fully compatible with the file formats used for all five drivers included in HASH v.1.2 (Fig. 5). In addition, SKHASH can also read the new polarity formats produced by the Northern California Earthquake Data Center (Northern California Earthquake Data Center [NCEDC], 2014) as well as a new “free form” format (i.e., the ‘SKHASH’ format). When using this free form format, users can provide a comma separated file with labeled columns in any order and can handle virtually any amount of whitespace or additional information.

User-defined variables are provided using a “control file,” a human-readable text file. Variable names precede the desired value, and these variables can be declared in any order. If a value is not provided in the control file, the default value is used. We have also attempted to make it as simple as possible to run the code in parallel and to take advantage of an API that calls a Fortran grid-search backend (both described later). These features are completely optional, and whether each feature used is controlled by a corresponding variable in the control file.

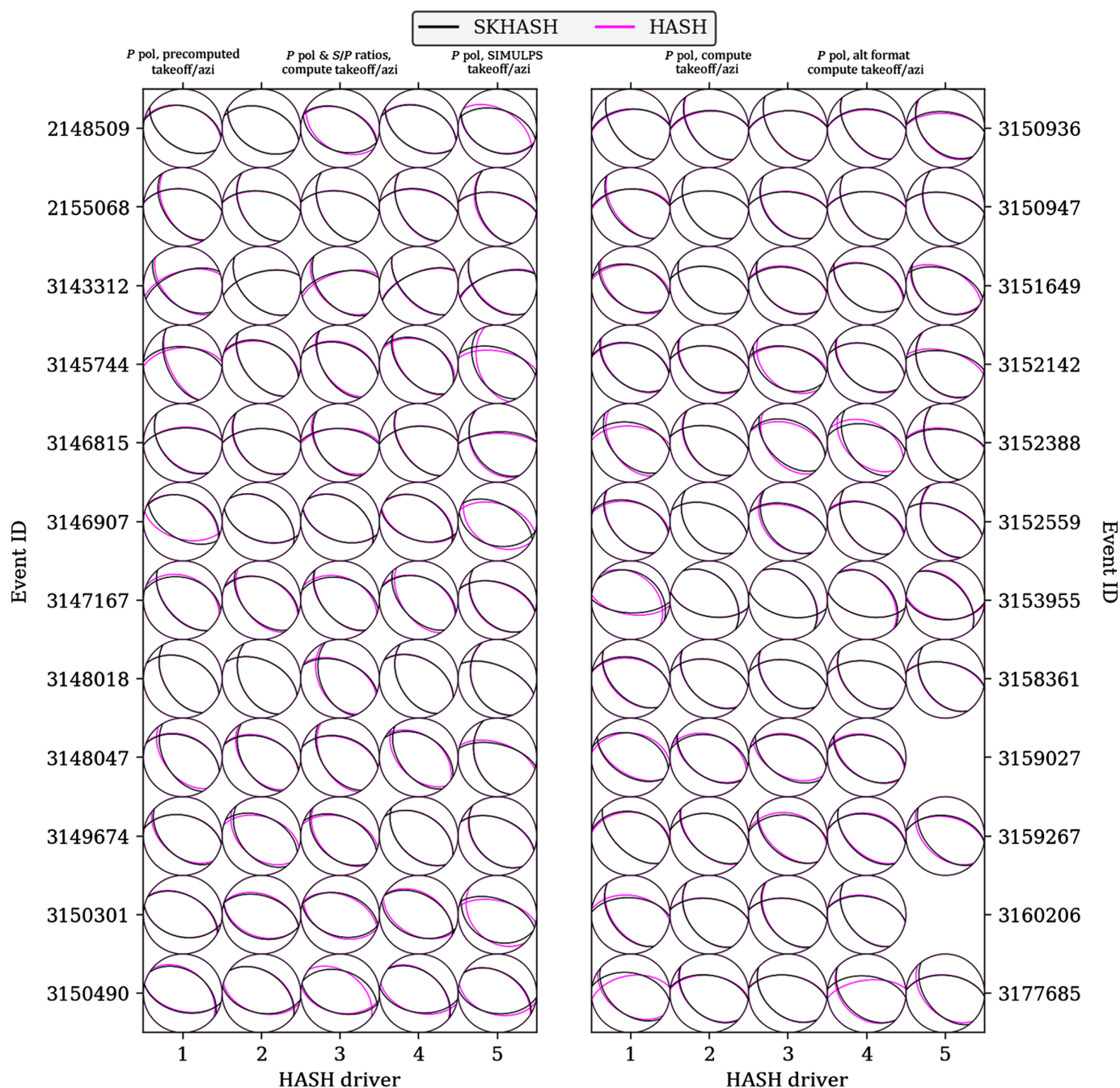
Efficiency

Python is largely an interpreted scripting language where converted byte code is executed in a virtual environment during execution. By comparison, Fortran is a language that requires code to be precompiled into a binary executable. As a result, faster runtimes for a variety of applications can be expected using Fortran than “pure” Python. To make the computation

Figure 4. Demonstration of composite focal mechanism solutions produced by SKHASH (Skoumal *et al.*, 2024) for two example families of earthquakes with (a) ~N49°E and (b) ~N11°W strikes (Shelly *et al.*, 2023; Skoumal, Shelly, and Hardebeck, 2023) that were part of an earthquake swarm that occurred in 2020 along the Maacama fault, California. The color version of this figure is available only in the electronic edition.

of mechanisms more efficient using Python in SKHASH, many operations are vectorized and the NumPy library is used, which utilizes precompiled C code. Most of the computation time is spent doing the grid search to find acceptable fault plane solutions. With SKHASH, the user has the option to use a Python C API that calls a Fortran grid-search subroutine. Provided a Fortran compiler is available on the user’s machine, this API is built automatically the first time the user requests the use of the Fortran backend. In addition, calculating focal mechanisms is an embarrassingly parallel problem and run-times can be greatly improved using parallel processing. If the user wishes, SKHASH can take advantage of Python’s multiprocessing module and run in parallel by simply specifying either a set number or all available Central Processing Units on their machine.

SKHASH runtimes to produce focal mechanisms depend on the user’s computer, whether the SKHASH Fortran grid search is used, and the complexity of focal mechanism problem. Depending on these factors, SKHASH may produce focal mechanisms with runtimes roughly equivalent to (or potentially quicker than) HASH (Fig. 6). For tasks that involve the computation of very large numbers of mechanisms or where near-real-time results are needed, HASH may still be the better suited tool provided only traditional *P*-wave polarities and *S/P* measurements are needed. For small to moderate tasks,



SKHASH and HASH runtimes may be virtually identical from a user's perspective. The flexibility of the inputs accepted by the SKHASH algorithm may simplify the process from formatting the input data allowing focal mechanisms to be computed easier and quicker.

Application

In our software package (see [Data and Resources](#)), we provide a user manual and examples of how to use SKHASH ([Skoumal et al., 2024](#)). The examples include (1) duplication of the five drivers used by HASH that demonstrate backward compatibility ("hash1-5"; Fig. 5), (2) synthetic *P*-wave polarities and *S/P* ratios that demonstrate the quality reporting produced by

Figure 5. Demonstrating the backward compatibility of our algorithm. Comparison of SKHASH (black; [Skoumal et al., 2024](#)) and HASH (magenta; [Hardebeck and Shearer, 2003](#)) focal mechanism solutions results for 24 earthquakes from the 1994 Northridge, California, sequence that were used in the five driver examples included in HASH v.1.2. As the input data and parameters are the same, the resulting solutions are virtually identical with subtle variations explained by the differing Monte Carlo trials. Note that two mechanisms are not produced using the inputs for HASH driver 5 as the azimuthal gap exceeds the user-defined value. The color version of this figure is available only in the electronic edition.

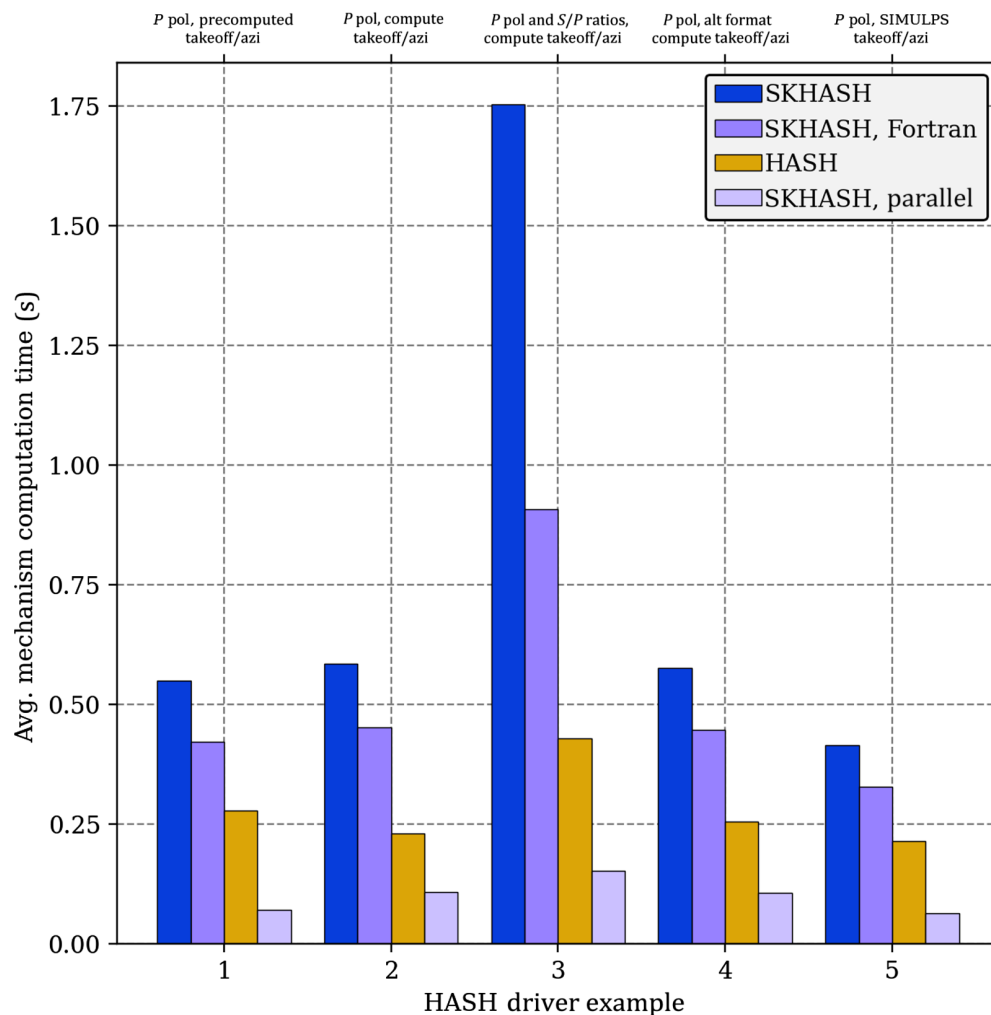


Figure 6. Runtimes of HASH v.1.2 using the five example drivers (“HASH”) compared with SKHASH (Skoumal *et al.*, 2024) using pure Python (“SKHASH”), the Python C API calling a Fortran grid-search subroutine (“SKHASH, Fortran”), and the Python C API run in parallel on 24 threads (“SKHASH, Parallel”). Runtimes were computed using 2xIntel Xeon Gold 5217 Processors. Each implementation was run 10 times, and the average system runtime to compute a mechanism are shown. Standard deviations of the runtimes were all <0.01 s. The color version of this figure is available only in the electronic edition.

SKHASH (“smile”; Fig. 3), and (3) composite mechanisms produced for the Maacama fault using correlation consensus (“maacama”; Fig. 4).

Conclusions

The number of open-source Python tools for seismological research have grown over the past couple of decades. Python packages, such as ObsPy (e.g., Beyreuther *et al.*, 2010), have offered seismologists the ability to conduct research and scientific support using streamlined workflows. Users familiar with Python may find the SKHASH code easier to understand, use, and modify for focal mechanism determination when compared with the preceding Fortran routines. Although SKHASH is not intended to be a complete replacement for all cases, we hope the opportunity to compute focal mechanisms in streamlined

environments will be appealing to many users. The improvements to the ease of use may appeal to scientists with limited scientific programming experience, and the capability of exploiting several recent advancements in focal mechanism determination may also attract researchers interested in the latest techniques. We plan to integrate future techniques into this SKHASH framework so that new methods are more accessible to the broader community, and we welcome all who are interested to join in its future development.

Data and Resources

The SKHASH software (Skoumal *et al.*, 2024; doi: 10.5066/P9MI5BUQ) is publicly available. Python (<https://www.python.org>), the NumPy package (<https://numpy.org>), and the pandas package (<https://pandas.pydata.org>) are all freely available. The HASH program (<https://www.usgs.gov/node/279393>) is also freely available. All websites were last accessed in February 2024.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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